

AROOJ FATIMA

AI Engineer | NLP & Generative AI | MLOps

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with of hands-on experience designing and deploying intelligent AI systems at production scale. Proven expertise in RAG pipelines, LLM fine-tuning, multi-agent architectures, and NLP. Delivered end-to-end solutions — from government AI programs to voice-enabled LMS assistants — combining research-driven methodology with practical MLOps deployment on AWS and Google Cloud.

CORE SKILLS

Languages	Python, SQL
AI / ML	NLP, Generative AI, LLMs (Gemini, OpenAI, Llama, Falcon), Machine Learning
Frameworks	TensorFlow, LangChain, LangGraph, CrewAI, Hugging Face, Google ADK
Tools	Git, Docker, Jira, Streamlit, Flask
Deployment	MLOps, AWS (EC2, ECR), Google Cloud, CI/CD (GitHub Actions)

PROFESSIONAL EXPERIENCE

AI Engineer | [Corvit Networks](#) Nov 2025 — Present

- Architected an intelligent career guidance bot for the **Digital Youth Hub**, integrating multi-platform data to deliver personalised career roadmaps and job matches; supplemented by an admin-side analytics engine surfacing sentiment trends and quality improvement reports.
- Engineered a **voice-enabled AI assistant** for a campus LMS, supporting natural-language commands across system modules (attendance, grading) with real-time insights on pending assignments and upcoming events.
- Built an end-to-end Sales Automation System for a networking firm, automating the full cycle from email query intake through SQR, quotation, proposal, and tender generation — cutting manual workforce requirements by **80%**.

Data Scientist | [DataWorld](#) | [Project Collaboration Program](#) Jan 2025 — Apr 2025

- Developed a Credit Card Time-Series forecasting model using Facebook Prophet, improving predictive accuracy on transactional data.
- Conducted FIFA dataset EDA on large-scale database records, engineering a match result predictor and defensive pressure classifier.
- Identified key cardiovascular risk indicators (cholesterol, BP, age) through statistical analysis of heart disease datasets.
- Designed and evaluated a Credit Card Fraud Detection pipeline, applying anomaly detection techniques to flag high-risk transactions.

TEACHING EXPERIENCE

Junior Lecturer | [University of Management & Technology \(UMT\)](#) Sep 2025 — Feb 2026

- Delivered lectures and lab sessions; developed teaching resources that improved student comprehension of core CS concepts.

Teaching Assistant — NLP, OS & DBMS | [UMT](#) Oct 2024 — Jul 2025

- Mentored students in NLP (NLTK, TensorFlow), OS process scheduling, and database design; reduced project revision cycles by providing structured code reviews and model debugging support.

KEY PROJECTS

LegalAI.Pak *Final Year Project — UMT*

- Designed a domain-specific legal RAG system for Pakistani law with automated ingestion (PDF/DOCX/TXT), FAISS + BM25 hybrid indexing, and reranking — achieving **85% Top-1 and 99% on Top-5 retrieval accuracy** and **MRR of 0.92**.
- Implemented citation verification (court match, year match, recall/precision) and an argument builder, validated by RAGAS-style metrics to ensure hallucination-free, trustworthy outputs for legal professionals.
- Engineered a full-stack legal platform with lawyer profile management, a case workspace (real-time communication, milestone tracking, secure document sharing), and a community Q&A forum.

Tech: Python, OpenAI API, LangChain, FAISS, Pinecone, Flask, ReactJS, ExpressJS, SQL

SmartShop.AI [Capstone](#) — [Kaggle GenAI Course](#)

- Built a multi-agent recommendation system with a 3-layer RAG pipeline (scraping → vector embedding → AI summarisation) to deliver sentiment-enriched product analysis, deployed with a Streamlit frontend and low-latency voice interface.

Tech: CrewAI, Gemini, Whisper, Render

Sonic Guide.AI [Hackathon](#) — [The Internet of Agents](#)

- Developed a multi-agent platform generating personalised, real-time audio tours for global locations by orchestrating specialised agents for architecture, history, culinary, and culture content.

Tech: Python, OpenAI Agents, GPT-4o, TTS, Streamlit

Medical Chatbot [Hackathon](#) — [GenAI Project](#)

- Deployed a production-grade medical chatbot on AWS (EC2/ECR) with Docker and GitHub Actions CI/CD, leveraging a curated medical knowledge base for preliminary diagnosis and responsible AI safeguards.

Tech: GPT-3.5, Pinecone, LangChain, Flask, AWS EC2/ECR, Docker, GitHub Actions

EDUCATION

Bachelor of Software Engineering (BSSE)

2021 — 2025

University of Management & Technology | **CGPA: 3.62 / 4.0**

CERTIFICATIONS

Generative AI Engineer — [Kaggle](#)

Google Data Scientist — [Udemy](#)

Certified Python Developer — [PFTP](#)

LSTWSTC'25 Speaker — [IEEE](#)

AWARDS & SCHOLARSHIPS

- Merit-Based Scholarship — **90% tuition fee waiver** awarded by UMT for outstanding academic performance.
- Dean's Award — Recognised for achieving **3.93 SGPA** in Fall 2023.